

Session 6

Fertility, Contraception, and Reproductive Health

Period and Cohort Fertility Measures, Proximate Determinants, Contraceptive Prevalence, and Unmet Need

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Learning Objectives

1

Distinguish fertility, fecundity, and natality

Use each concept and its correct measures precisely, and identify the standard data sources behind fertility statistics.

2

Compute the full family of fertility measures

Child-Woman Ratio, Crude Birth Rate, General Fertility Rate, Age-Specific Fertility Rates, and the Total Fertility Rate, with two full worked examples.

3

Compare major theories of fertility transition

Classic demographic transition theory, the neoclassical microeconomic theory of fertility, and regulatory/behavioral theories of fertility decline.

4

Apply the Bongaarts proximate determinants framework

Explain how marriage, contraception, induced abortion, and postpartum infecundability jointly determine observed fertility.

5

Evaluate contraceptive prevalence and unmet need

Interpret these indicators using real, current country data, including Thailand and other very-low-fertility settings.

6

Analyze real fertility policy and decline scenarios in groups

Apply this session's theories and frameworks to real countries and real pro-natalist policies in two structured group-work activities.

AGENDA

Session Structure (180 minutes)

Block 1

Concepts, Measurement & Two Worked Examples

Definitions & data sources; CWR, CBR, GFR, ASFR, TFR; a full worked TFR calculation plus a second real-data worked example.

Break

15-minute short break

Before the theory and framework block.

Block 2

Theories & Group Work 1

Demographic transition theory, microeconomic theories of fertility; a real-country group-work analysis.

Block 3

Proximate Determinants, Contraception & Unmet Need

The Bongaarts framework, contraceptive method mix, unmet need, and real-world cases.

Block 4

Below-Replacement Fertility & Group Work 2

Global lowest-low fertility patterns; a real pro-natalist policy evaluation; practice exercises and discussion.

Why This Matters Right Now: Thailand's Births Have Collapsed

462,240

Births registered in Thailand, 2024

571,646

Deaths registered in Thailand, 2024

≈1.0

Estimated Thai total fertility rate, 2024

For the first time since 1949, annual births in Thailand fell below 500,000 — while deaths exceeded 570,000, meaning Thailand's population is now shrinking through natural decrease. Understanding why requires exactly the measurement toolkit and theory this session builds: from the Crude Birth Rate you will compute by hand today, to the proximate-determinants and transition theories explaining the decline.

Source: Khaosod English. (2025, January 17). Thailand's population drops below 66 million as births hit 75-year low.
<https://www.khaosodenglish.com/featured/2025/01/17/thailands-population-drops-below-66-million-as-births-hit-75-year-low/>

BLOCK 1

Concepts, Measurement & a Worked Example

Defining fertility precisely, the standard data sources, and building every core fertility measure from raw data.

Fertility, Fecundity, and Natality: Three Different Concepts

Fertility

The actual reproductive performance of an individual or population — the number of live births that actually occur. This is what fertility RATES measure.

Fecundity

The physiological CAPACITY to bear children — a potential, not an outcome. A woman can be highly fecund yet have zero fertility (by choice, circumstance, or lack of exposure to intercourse).

Natality

A broader, population-level term for the frequency of births in a population — often used loosely as a synonym for fertility, but properly refers to the birth phenomenon itself.

Why the distinction matters

Proximate-determinants analysis (Block 3) exists precisely because the gap between fecundity (biological maximum) and fertility (observed outcome) is explained by behavior — marriage, contraception, abortion, breastfeeding.

Where Fertility Data Come From

1

Civil (vital) registration systems

Continuous recording of every birth as it occurs — the gold standard when coverage and quality are high (as in Thailand's Department of Provincial Administration registration system), enabling annual CBR, GFR, ASFR, and TFR.

2

Population censuses

Periodic (typically decennial) full enumeration; often includes retrospective fertility questions ("children ever born," "births in the last 12 months") that allow indirect estimation where registration is weak.

3

Sample surveys

Demographic and Health Surveys (DHS), Multiple Indicator Cluster Surveys (MICS), and national surveys collect detailed birth histories, enabling cohort fertility analysis, contraceptive prevalence, and unmet need — not just period rates.

Choosing a source

Registration data give the most timely, precise period rates; surveys are essential for behavioral indicators (contraception, unmet need, proximate determinants) that registration systems do not capture.

Common Misconceptions About Fertility Measures



"CBR and TFR measure the same thing."

They do not. CBR divides births by the ENTIRE population and is heavily distorted by age-sex structure. TFR is built from age-specific rates among women of reproductive age and is comparable across populations with very different structures. A country can have a low CBR but a moderate TFR simply because it has few women of childbearing age relative to its total population.



"A falling period TFR always means women want fewer children."

Not necessarily — this conflates tempo and quantum. If women are simply postponing births to later ages (a timing/tempo shift), period TFR falls temporarily even though completed cohort fertility (the quantum) may barely change. Always ask whether mean age at childbearing is rising before concluding that desired family size has fallen.



"Replacement-level fertility is always exactly 2.0 or exactly 2.1."

2.1 is a common low-mortality benchmark, but the true replacement level depends on the sex ratio at birth and, especially, on mortality before the end of the reproductive years. Where child and young-adult mortality remain substantial, replacement-level TFR can be considerably higher — up to 2.5–3.3 in some high-mortality settings.

Child-Woman Ratio (CWR): A Census-Based Proxy

$$\text{CWR} = (\text{Children aged 0-4}) / (\text{Women aged 15-49}) \times 1,000$$

Useful when no birth registration exists

Computable from a single census age structure alone — no vital registration required, which is why it is still used for historical and low-data-quality settings.

Key limitations

Distorted by child mortality (undercounts fertility where infant/child mortality is high) and by migration of children or women of reproductive age between the census and the reference period.

Not age-specific

Gives one crude population-level ratio — cannot show which reproductive ages are driving fertility, unlike ASFR.

Crude Birth Rate (CBR)

$$\text{CBR} = (\text{Total live births in a year}) / (\text{Mid-year total population}) \times 1,000$$

Easy to compute, easy to misinterpret

The denominator is the ENTIRE population — men, children, the elderly — not just women capable of childbearing.

Major limitation: sensitive to age-sex structure

Two populations with identical age-specific fertility behavior can show very different CBRs simply because one has a younger, more female-heavy population.

Why demographers move beyond it

This structure-sensitivity is exactly why GFR, ASFR, and TFR were developed — each restricts or standardizes the denominator to isolate actual reproductive behavior.

General Fertility Rate (GFR)

$$\text{GFR} = (\text{Total live births in a year}) / (\text{Mid-year women aged 15-49}) \times 1,000$$

A real improvement over CBR

Restricting the denominator to women of reproductive age (15-49) removes the distortion from the number of men, children, and the elderly in the population.

Still not age-specific

Two populations with identical GFR can have very different underlying age patterns — one concentrated at ages 20-24, another spread evenly — GFR cannot distinguish them.

Worked example (preview)

Population 1,000,000; women 15-49 = 260,000; births = 14,000/year → CBR = 14.0 per 1,000 total population; GFR = $(14,000/260,000) \times 1,000 = 53.8$ per 1,000 women 15-49.

Age-Specific Fertility Rate (ASFR)

$${}_nF_x = (\text{Births to women aged } x \text{ to } x+n) / (\text{Mid-year women aged } x \text{ to } x+n) \times 1,000$$

Computed separately for each age group

Typically 5-year groups: 15-19, 20-24, 25-29, ... 45-49 — this is what lets us see the SHAPE of the fertility schedule, not just its level.

The most analytically rich fertility measure

ASFR reveals timing (early vs. late childbearing), which CBR and GFR completely obscure — essential for comparing fertility postponement across cohorts or countries.

Feeds directly into TFR

The Total Fertility Rate (next slide) is simply built by summing ASFRs across all reproductive ages.

Total Fertility Rate (TFR)

$$\text{TFR} = n \times \sum(nFx) / 1,000 \quad [\text{sum of all ASFRs, } \times \text{ age-interval width, typically } n = 5]$$

A synthetic-cohort measure, like e_0 in a life table

TFR answers: "If a hypothetical woman experienced today's observed ASFR schedule at every age throughout her reproductive life, how many children would she have?"

Replacement level ≈ 2.1

The TFR at which a population, in the absence of migration, exactly replaces itself across generations (slightly above 2.0 to offset mortality before reproductive age).

Period vs. cohort TFR

Period TFR (computed here) can be distorted by "tempo effects" — if women are simply postponing births, period TFR falls temporarily even if completed (cohort) family size does not change as much.

Period vs. Cohort Fertility, and Mean Age at Childbearing

Period TFR

A snapshot: applies one calendar year's ASFR schedule to a synthetic cohort. Fast to compute, immediately available — but sensitive to short-term timing shifts.

Cohort (completed) fertility

The actual average number of children born to women of a real birth cohort by the end of their reproductive lives. Only fully known decades later.

Tempo effects

When women delay childbearing (rising mean age at first birth), period TFR can fall even if cohort fertility barely changes — a key reason to interpret sudden period-TFR declines cautiously.

$$\text{MAC} = \frac{\sum(x_{\text{mid}} \times {}_nF_x)}{\sum({}_nF_x)}$$

Mean age at childbearing (MAC), where x_{mid} is the mid-point of each age group — summarizes WHEN, on average, childbearing occurs within the ASFR schedule.

Worked Example: Age-Specific Fertility Rates

Age group	Births to women in group	Women in group (mid-year)	nFx (per 1,000)
15–19	960	120,000	8.0
20–24	5,175	115,000	45.0
25–29	8,970	115,000	78.0
30–34	6,944	112,000	62.0
35–39	3,024	108,000	28.0
40–44	700	100,000	7.0
45–49	46	92,000	0.5

Illustrative population, loosely modeled on a low-fertility Southeast Asian setting. $nFx = (\text{births/women}) \times 1,000$, e.g. age 20–24: $(5,175/115,000) \times 1,000 = 45.0$.

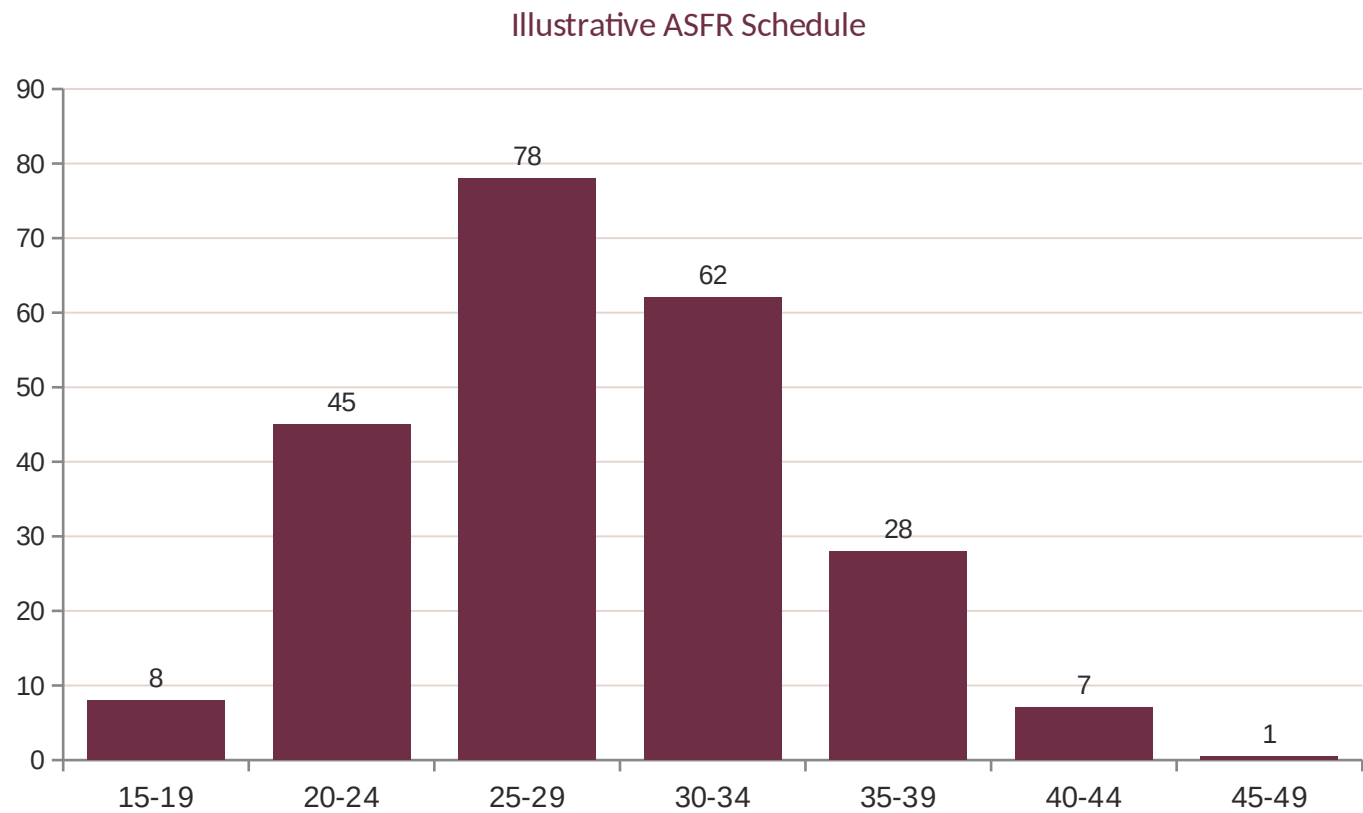
Worked Example: Computing the Total Fertility Rate

Age group	nFx	Contribution = $(nFx \times 5) / 1,000$
15–19	8.0	0.0400
20–24	45.0	0.2250
25–29	78.0	0.3900
30–34	62.0	0.3100
35–39	28.0	0.1400
40–44	7.0	0.0350
45–49	0.5	0.0025

$TFR = 5 \times \Sigma(nFx) / 1,000 = 5 \times 228.5 / 1,000 = 1.14$ children per woman

Well below the replacement level of 2.1 — consistent with the Thailand-like pattern in Slide 4.

The Fertility Schedule: Where Childbearing Concentrates



Peak at 25-29

Fertility is concentrated in the mid-to-late twenties in this schedule — the mode of the age pattern.

Right-skewed decline

The fall-off after age 30 is steady but slower than the rise before 25.

Policy relevance

A schedule shifting rightward over time (peak moving toward 30-34) is the signature of fertility postponement, seen across urban East and Southeast Asia.

A Real ASFR Schedule: The Philippines, 2022

Everything so far used an illustrative population. Now we compute the same measures from real, published survey data — the 2022 Philippine National Demographic and Health Survey (NDHS) — and check our arithmetic against the officially published TFR.

Age group	ASFR (births per 1,000 women), 2022 NDHS
15–19	25
20–24	84
25–29	105
30–34	95
35–39	58
40–44	21
45–49	2

Source: Philippine Statistics Authority. (2022, November 13). Total fertility rate declined from 2.7 in 2017 to 1.9 in 2022 [Preliminary results, 2022 NDHS]. <https://psa.gov.ph/content/total-fertility-rate-declined-27-2017-19-2022>

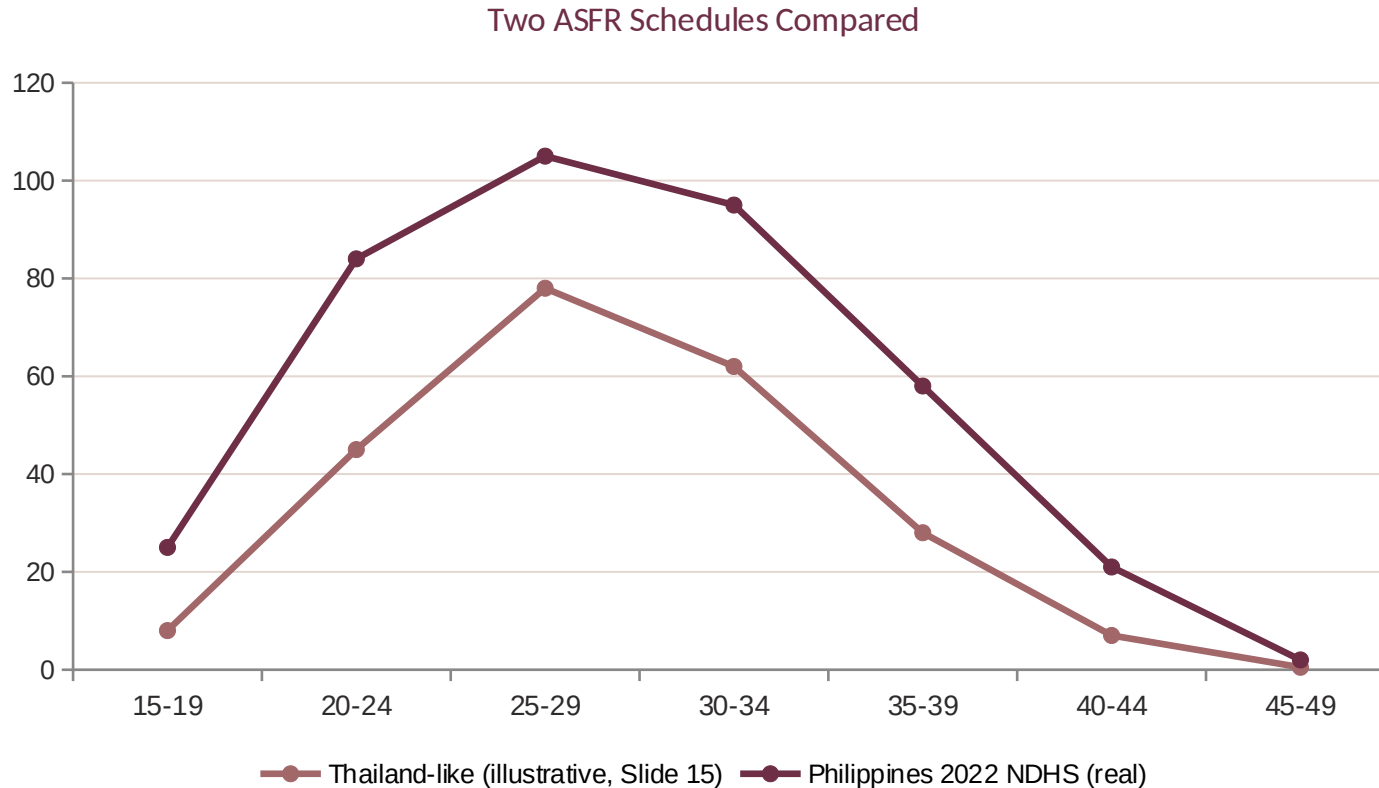
Summing the Philippines' ASFR Schedule to TFR

Age group	ASFR	Contribution = (ASFR × 5)/1,000
15–19	25	0.125
20–24	84	0.420
25–29	105	0.525
30–34	95	0.475
35–39	58	0.290
40–44	21	0.105
45–49	2	0.010

Computed TFR = $5 \times \Sigma(\text{ASFR})/1,000 = 5 \times 390/1,000 = 1.95 \approx 1.9$ children per woman

This matches the PSA's officially published 2022 NDHS TFR of 1.9 almost exactly (the small gap is rounding in the published per-1,000 rates) — a genuine consistency check on the ASFR-to-TFR method.

Comparing Two Real Fertility Schedules: Thailand vs. the Philippines



Same peak age, very different level

Both schedules peak at 25–29, but the Philippines' rate at every age is far higher — its TFR (1.9) is nearly double Thailand's (≈ 1.0).

Mean age at childbearing

MAC = $\Sigma(x_{\text{mid}} \times \text{ASFR}) / \Sigma(\text{ASFR}) \approx 29.4$ years for the Philippines schedule — try computing this yourself as practice.

Both below replacement

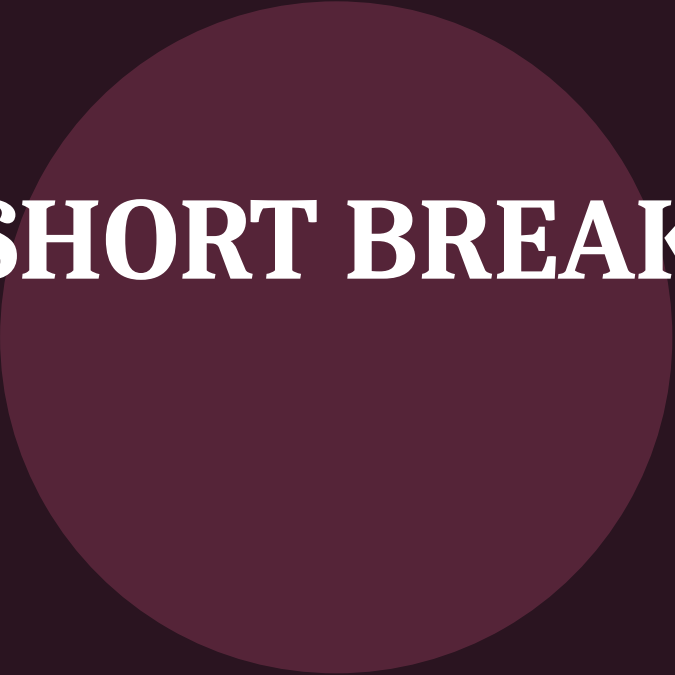
Even the higher of the two (1.9) sits below the 2.1 replacement benchmark — illustrating how widespread below-replacement fertility has become in Asia.

Comparing the Three Core Fertility Rates

Measure	Denominator	Main strength	Main limitation
CBR	Entire population	Simple; needs only total births and population	Highly sensitive to age-sex structure
GFR	Women aged 15–49	Removes distortion from men, children, elderly	Still not age-specific — hides timing patterns
TFR (from ASFR)	Women in each 5-yr age group	Standardized; comparable across populations and time	Period version subject to tempo distortion

Rule of thumb: always prefer TFR for cross-country or cross-time comparison; use CBR only when only aggregate vital-registration totals are available.

The computational lab



SHORT BREAK

BLOCK 2

Theories of Fertility Transition

Why fertility falls: from the classic demographic transition to microeconomic and behavioral explanations.

Classic Demographic Transition Theory

The core sequence

Pre-transition: high mortality, high fertility, slow growth. Stage 2: mortality falls first (sanitation, medicine), fertility stays high → rapid growth. Stage 3: fertility falls toward mortality. Stage 4: low mortality, low fertility, slow/negative growth.

Why fertility eventually falls

Original theory (Notestein and others) emphasized urbanization, industrialization, rising costs of children, declining infant mortality, and the shift from extended to nuclear family systems.

Strengths and critiques

Fits the historical European experience reasonably well but has been criticized for treating fertility decline as an almost automatic by-product of "modernization" rather than a set of specific, contestable behavioral choices.

Neoclassical Microeconomic Theory of Fertility (Becker)

Children as a "durable good"

Households choose quantity (n) and quality (q) of children subject to a budget and time constraint, maximizing utility $U(n, q, \text{other goods})$.

The quantity–quality trade-off

As household income and the opportunity cost of parental (especially maternal) time rise, parents rationally substitute toward fewer children with more investment per child — "quality" becomes relatively cheaper than "quantity."

Rising cost of children drives fertility down

Compulsory schooling, child labor laws, and rising female wages all raise the direct and opportunity cost of childbearing — consistent with fertility decline alongside female education and labor-force gains.

Source: Becker, G. S. (1960). An economic analysis of fertility. In Demographic and economic change in developed countries (pp. 209–240). Princeton University Press for NBER.

Regulatory (Behavioral) Microeconomic Theory of Fertility (Easterlin)

Three components of observed fertility

(1) demand for children, (2) supply of children (natural fertility net of infant/child mortality), and (3) the costs of fertility regulation (access to and cost of contraception).

Adds "relative income" and aspirations

Fertility desires are shaped by comparing material aspirations (formed in childhood) against current earning potential — cohorts facing a favorable ratio tend toward higher fertility, and vice versa.

Why "regulatory": the missing piece in Becker's model

Even where demand for children is low, fertility only falls in practice if effective, low-cost fertility regulation is available — the bridge to the Bongaarts proximate-determinants framework in Block 3.

Source: Easterlin, R. A. (1975). An economic framework for fertility analysis. Studies in Family Planning, 6(3), 54–63.

Comparing the Three Theoretical Traditions

Theory	Core driver of decline	Best explains
Classic demographic transition	Modernization: falling mortality, urbanization, industrialization	Long-run, cross-country historical patterns
Neoclassical (Becker)	Rising cost of children and quantity-quality trade-off	Why richer, more educated households have fewer, more "invested-in" children
Regulatory/behavioral (Easterlin)	Relative income/aspirations plus access to fertility regulation	Short/medium-run fluctuations and WHY desired declines translate into actual decline

These theories are complementary, not competing: transition theory explains the long arc, Becker explains household-level resource allocation, and Easterlin explains why the translation from desire to outcome depends on access to regulation.

Order of TFR, highest to lowest (Nigeria → Philippines → Thailand → South Korea)

Four Real Countries, Four Different Fertility Stories

Group A — Thailand

TFR $\approx$ 0.93 (2025, Mahidol Population Gazette). Women 15-49: 15.6M. CPR 73%, but marriage rates are falling sharply. Below-replacement for over 30 years.

Institute for Population and Social Research, Mahidol University (2025). Mahidol Population Gazette (Vol. 34).

Group B — South Korea

TFR = 0.75 (2024, Statistics Korea, first uptick in a decade after a record-low 0.72 in 2023). Extreme housing and private-education costs; long working hours.

The Korea Herald (2025). Births in Korea rebound for 1st time in nearly a decade in 2024.

Group C — Philippines

TFR = 1.9 (2022 NDHS), down from 2.7 in 2017 — a rapid decline, but still the highest of the four, and just below replacement.

Philippine Statistics Authority (2022). Total fertility rate declined from 2.7 in 2017 to 1.9 in 2022.

Group D — Nigeria

TFR = 4.8 (2024 NDHS), down from 5.3 in 2018. Modern contraceptive prevalence only 15%; over half of demand for family planning remains unmet in most states.

National Population Commission & ICF (2025). Nigeria Demographic and Health Survey 2024, key indicators report.

BLOCK 3

Proximate Determinants, Contraception & Unmet Need

The Bongaarts framework linking fecundity to observed fertility, and the real state of family planning worldwide.

The Bongaarts Framework: The Big Idea First

Every population has a biological ceiling on fertility — total fecundity (TF), roughly 15.3 children per woman if nothing at all inhibited childbearing. No real population gets anywhere near that number. Bongaarts (1978) asked a simple question:

"What are the specific, measurable behaviors that stand between biological potential ($TF \approx 15.3$) and observed fertility (TFR), and how much does each one subtract?"

His answer: just four proximate determinants explain almost all cross-population variation in fertility. Each is measured as an INDEX from 0 (fully inhibiting — fertility driven to zero) to 1 (no inhibiting effect at all).

C_m

Marriage / union

C_c

Contraception

C_a

Induced abortion

C_i

Postpartum infecundability

Index of Marriage / Union (Cm)

Cm

Cm falls as fewer women of reproductive age are married or in a sexual union — because in most societies, most childbearing occurs within union.

What drives Cm down

Later average age at first marriage, rising proportions never-married, and rising divorce/separation all reduce the time women spend exposed to the risk of pregnancy within union.

Real-world driver

In contemporary Thailand and South Korea, Cm — not contraception — is now the dominant driver of low fertility: marriage itself is being delayed or foregone, so very little childbearing opportunity exists in the first place, regardless of contraceptive access.

Index of Contraception (Cc)



Cc falls as contraceptive prevalence rises and as the methods used become more effective (e.g., sterilization and IUDs inhibit fertility far more than periodic abstinence).

What drives Cc down

Expanding family-planning programs, health-system integration of contraceptive services, and a shift in method mix toward long-acting, highly effective modern methods.

Real-world driver

Thailand's own history is the textbook case: contraceptive prevalence rose from 14.8% (1970) to roughly 79% by the early 2010s — one of Asia's fastest family-planning transitions, driven initially by a strong national program.

Index of Induced Abortion (Ca)

Ca

Ca falls as the rate of induced abortion rises, since each induced abortion removes a pregnancy that would otherwise have resulted in a live birth.

What drives Ca down

Legal access to safe abortion, but also high rates of induced abortion under restrictive legal regimes (unsafe, clandestine abortion still counts toward Ca in the demographic sense, even though it carries severe health risks.)

Real-world driver

During East Asia's rapid fertility decline (1960s–1980s), several countries — South Korea among them — relied heavily on induced abortion alongside rising contraceptive use; Ca and Cc fell together rather than sequentially.

Index of Postpartum Infecundability (Ci)

Ci

Ci falls as the average duration of postpartum infecundability lengthens — mainly driven by longer, more intensive breastfeeding, which suppresses ovulation.

What drives Ci down (i.e., MORE inhibition)

Longer average breastfeeding duration and postpartum abstinence norms extend the biological "resting" period after each birth, naturally spacing births further apart even without any contraceptive use.

Real-world driver

In historical high-fertility, pre-transition sub-Saharan Africa, long breastfeeding durations (low Ci) were often the single biggest natural brake on fertility — long before modern contraception was widely available.

The Bongaarts Multiplicative Model

$$TFR = C_m \times C_c \times C_a \times C_i \times TF$$

Each index ranges from 0 (maximum inhibition) to 1 (no inhibiting effect). Multiplying all four together against total fecundity (TF ≈ 15.3) reconstructs the observed TFR.

- Total fecundity in biology is the total physiological reproductive capacity or potential output of an individual organism (typically a female) expressed as the number of eggs, seeds, or offspring produced during a lifetime or a single reproductive cycle.
- Often confused with total fertility rate (TFR)—which in demography measures the average number of children born per woman in a population—fecundity strictly addresses biological potential rather than actual social or demographic output

Index	Falls (toward 0) when...	Typically the dominant lever when...
C _m	Fewer women are married/in union during reproductive ages	Marriage is being delayed or foregone (contemporary East/SE Asia)
C _c	Contraceptive prevalence and effectiveness rise	A country is mid-transition with rising family-planning access
C _a	Induced abortion rates rise	Legal/clandestine abortion substitutes for contraception
C _i	Breastfeeding/postpartum abstinence durations lengthen	Pre-transition, high-fertility, low-contraception settings

In-Class Discussion: Which Determinant Dominates?

Historical sub-Saharan Africa (pre-transition)

High fertility sustained mainly by near-universal early marriage (high C_m) and very low contraceptive use (high C_c) — discuss which index would be closest to 1.

East Asia during rapid fertility decline (1960s–1980s)

Fertility fell rapidly through rising contraceptive prevalence AND, in several countries, high induced-abortion rates — discuss the relative roles of C_c and C_a .

Contemporary Thailand and South Korea (today)

Very low C_m (late/declining marriage) is now the dominant factor — discuss why this differs from the classic "contraception-driven" transition narrative.

Your task: for each scenario, identify which index (C_m , C_c , C_a , C_i) is likely closest to 0 (most inhibiting) and justify your answer using the definitions above.

Contraceptive Prevalence and Unmet Need: Definitions

Contraceptive Prevalence Rate (CPR)

The percentage of women of reproductive age (often married/in-union, 15–49) currently using any method of contraception, modern or traditional.

Modern CPR (mCPR)

Restricts CPR to modern, clinically effective methods (pill, IUD, implant, injectable, sterilization, condom) — the indicator most used in international family-planning tracking (e.g., FP2030).

Unmet need for family planning

Fecund, sexually active women who wish to stop or delay childbearing but are not using any method of contraception — the gap between reproductive intentions and contraceptive behavior.

Demand satisfied by modern methods

The share of total demand for family planning (current modern users + unmet need) that is actually met with modern methods — a core SDG 3.7 indicator.

Method Mix Matters: Modern vs. Traditional Contraception

A country's CPR can look similar on paper while its Bongaarts Cc index (and its effectiveness against unintended pregnancy) differs enormously — because the MIX of methods, not just the headline prevalence rate, determines how much protection is actually delivered.

Modern methods

Pill, IUD, implant, injectable, male/female sterilization, condom. Typical use-failure rates range from under 1% (sterilization, IUD, implant) to about 7-13% (pill, condom) — still far more effective than traditional methods.

Traditional methods

Periodic abstinence (rhythm/calendar method) and withdrawal. Typical use-failure rates are much higher — often 13-20%+ per year — meaning a given CPR built on traditional methods buys much less real fertility inhibition.

Thailand's method mix in practice

Thailand's contraceptive method mix is overwhelmingly modern — female sterilization and the oral pill have historically dominated, with injectables and IUDs also widely used through the public health system; traditional methods account for only a small residual share. This is a key reason Thailand's Cc fell so far, so fast — high prevalence AND high effectiveness, both moving together.

Sources: Panichkriangkrai et al. (2020). *Universal access to sexual and reproductive health services in Thailand*. *Sexual and Reproductive Health Matters*, 28(2); Multiple Indicator Cluster Survey (MICS) 2022, Thailand, NSO & UNICEF.

Thailand: A Family-Planning Success Story, With Room Left to Close

Dramatic long-run rise in contraceptive use

CPR rose from just 14.8% in 1970 to roughly 79% by the early 2010s, one of the fastest and most complete family-planning transitions in Asia, underpinned by Thailand's universal health coverage since 2002.

Unmet need has fallen but not disappeared

Unmet need among women aged 15-49 fell from 11.7% in 1987 to about 6.2% in 2016 — real progress, but not yet zero, particularly among adolescents and some rural populations.

79%

Contraceptive prevalence rate, Thailand, all methods
(women 15-49)

Link to the Bongaarts framework

This is a textbook case of Cc falling sharply over five decades, becoming the dominant proximate determinant behind Thailand's fertility decline to today's sub-replacement level.

Source: Panichkriangkrai, W. et al. (2020). Universal access to sexual and reproductive health services in Thailand. *Sexual and Reproductive Health Matters*, 28(2). <https://doi.org/10.1080/26410397.2020.1805842>

Unmet Need Remains a Global Reality, Not Just a Low-Income-Country Problem

A persistent, worldwide gap

Globally, roughly one in ten married or in-union women of reproductive age has an unmet need for family planning.

Highly uneven across regions

As of 2019, 42 countries — 23 of them in sub-Saharan Africa — still had less than half of demand for family planning satisfied by modern methods. Nigeria's 2024 NDHS shows the same pattern still holding: modern CPR only 15%, and no state has yet met the 10% unmet-need target.

~1 in 10

Married/in-union women worldwide with unmet need for family planning

BLOCK 4

Below-Replacement Fertility: Global Patterns

From replacement level to the new demographic reality of very-low fertility across East and Southeast Asia — and beyond.

Replacement-Level Fertility

Replacement-level TFR $\approx$ 2.1 children per woman (low-mortality settings; up to ~2.5-3.3 where child mortality remains substantial)

Why not exactly 2.0?

The extra 0.1 accounts for the small proportion of girls who will not survive to, or through, their own reproductive years, and for a slightly higher number of male births per female birth (sex ratio at birth $\approx$ 105:100).

Below-replacement $\neq$ immediate population decline

Population momentum (a young age structure inherited from past higher fertility) can sustain population growth for one to two more generations even after TFR falls below 2.1.

A growing share of humanity

Over half of the world's population now lives in a country or area with below-replacement fertility — no longer a phenomenon confined to a handful of high-income countries.

The New Global Fertility Landscape

2.1

Replacement-level TFR (benchmark)

2.3

Sub-Saharan Africa, approx. regional TFR (still above replacement)

<1.0

Several East/SE Asian TFRs, 2024-2025 (South Korea, Singapore)

The global fertility picture is now sharply bimodal: sub-Saharan Africa remains the last region with fertility clearly above replacement, while an expanding cluster of East and Southeast Asian countries — and a growing number of European ones — now sit not just below replacement, but below 1.5, a level demographers term "lowest-low" fertility.

Consequences of Sustained Below-Replacement Fertility

Accelerated population ageing

Fewer births directly raises the old-age dependency ratio — connecting directly back to the prospective-ageing measures from Session 5.

Shrinking future labor force

Smaller birth cohorts today become smaller working-age cohorts 15-64 years later, with direct implications for economic growth, pension systems, and healthcare financing.

Limits of pro-natalist policy

Cash incentives, parental leave, and childcare subsidies (tried extensively in Hungary, South Korea, Japan, and Singapore) have so far produced only modest, often temporary, effects on TFR.

Migration as a partial offset

Several very-low-fertility countries increasingly rely on immigration to stabilize the working-age population — a theme picked up in the next session on migration.

Four Countries, One Pattern: East/Southeast Asia's Fertility Collapse

Country	Most recent TFR	Year	Note
South Korea	0.75	2024	World's lowest major-economy TFR; small rebound from 0.72
Singapore	0.97 (0.87 est.)	2024 (2025 est.)	First fell below 1.0 in 2023; continuing to decline
Japan	1.15	2024	Ninth consecutive annual record low; births below 700,000
Thailand	0.93	2025	Mahidol Population Gazette; births below 500,000 since 2024

All four are members of the "lowest-low fertility" club (TFR < 1.3), despite very different welfare-state models, housing markets, and pro-natalist policies — a strong signal that these theories (Blocks 2-3) must operate jointly, not singly.

GROUP WORK • ASSIGNED POLICY DATA

Three Real, Currently Active Fertility Policies

15 min prep + 3 groups × 5 min report-back + 5-10 min synthesis discussion

Group 1 — Hungary's Pro-Natalist Tax & Loan Package

~5% of GDP/year since 2010: lifetime income-tax exemption for mothers of 4+ children, subsidized loans with per-child debt forgiveness, housing subsidies. Result: TFR fell to 1.39 in 2024 (from 1.51 in 2023); births down 9.1%.

N-IUSSP (2024); KSH (Hungarian Central Statistical Office), 2024 data.

Group 2 — Singapore's Baby Bonus & Marriage/Parenthood Package

Cash gifts (up to S\$11,000 per child), Child Development Co-Savings accounts, and paid parental leave, expanded repeatedly since 2001. Result: TFR still fell to a record low (0.97 in 2024, 0.87 est. 2025).

Bloomberg (2026); Singapore Department of Statistics, Population in Brief.

Group 3 — Thailand's Integrated National Strategic Response

Rather than cash incentives, Thailand's 2025 national strategy emphasizes active aging, human-capital investment, and skilled immigration — treating low fertility as structural, not solved by pro-natalist payments alone.

Chulalongkorn University (2025). Integrated National Strategic Plan.

1. Which proximate determinant(s) is this policy actually trying to shift (Bongaarts' Cm, Cc, Ca, Ci), and does the outcome data suggest it worked?
2. Given the outcome, what does this case suggest about the limits (or conditions for success) of this policy approach — cash/tax incentives vs. structural investment — for raising fertility?
3. If you were advising this government next year, what one change would you recommend to the policy design, and why?

Key Takeaways

1

Measure fertility with the right tool for the question

CBR for a quick aggregate figure, GFR to remove population-structure distortion, ASFR/TFR for rigorous, comparable, age-detailed analysis — and always check your arithmetic against a real published TFR, as in Worked Example 2.

2

No single theory fully explains fertility decline

Classic transition theory, Becker's quantity-quality model, and Easterlin's regulatory/aspirations framework are complementary lenses, not competitors.

3

The Bongaarts framework is the analytical bridge

It connects biological fecundity to observed fertility through four measurable, policy-relevant proximate determinants — marriage, contraception, abortion, postpartum infecundability.

4

Below-replacement fertility is now a global-majority reality

From Thailand to Hungary to South Korea, pro-natalist policy has so far struggled to reverse it — reshaping population ageing, labor markets, and family-planning priorities worldwide.

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Practice Exercises (1 of 3) · Conceptual

15 questions across the next three slides — a mix of conceptual, calculation, and applied questions. Bring your worked answers to the next session for discussion.

- Q1.** Explain the difference between fertility, fecundity, and natality, giving one example situation where a woman has high fecundity but zero fertility.
- Q2.** Why is the Crude Birth Rate considered a poor measure for comparing fertility across two countries with very different age-sex structures? Name the measure that best corrects for this.
- Q3.** Explain, using the Bongaarts framework, why a country could have a low unmet need for contraception (like Thailand) yet still have a very low TFR.
- Q4.** Compare the neoclassical (Becker) and regulatory (Easterlin) theories of fertility: what does each add that the other lacks?
- Q5.** What is a "tempo effect," and why can it cause period TFR to understate long-run cohort fertility? Use Hungary's 2024 experience (Slide 50) in your answer.
- Q6.** State one key misconception about fertility measurement (Slide 8) in your own words, and explain why it is incorrect.

Practice Exercises (2 of 3) · Calculation

Q7. A population of 2,400,000 records 32,000 births in a year, of which the population of women aged 15-49 is 650,000. Compute the CBR and the GFR.

Q8. Given the following ASFRs (per 1,000 women): 15-19: 12; 20-24: 60; 25-29: 95; 30-34: 70; 35-39: 30; 40-44: 6; 45-49: 0.3 — compute the TFR.

Q9. A census records 18,000 children aged 0-4 and 95,000 women aged 15-49 in a district. Compute the Child-Woman Ratio. State one reason this might understate true fertility.

Q10. If $C_m = 0.65$, $C_c = 0.55$, $C_a = 0.98$, $C_i = 0.85$, and total fecundity $TF = 15.3$, use the Bongaarts model to estimate the implied TFR.

Q11. Using the Philippines ASFR schedule from Worked Example 2 (Slide 18), compute the Mean Age at Childbearing (MAC). Use age-group midpoints (17, 22, 27, 32, 37, 42, 47).

Q12. Thailand's CPR is roughly 73% and unmet need is low. Using the definitions from Slide 41, calculate the approximate percentage of total demand for family planning that is satisfied if unmet need is 5%. (Hint: total demand = current use + unmet need.)

Practice Exercises (3 of 3) · Applied

Q13. Using the comparative table on Slide 49, rank Thailand, Japan, South Korea, and Singapore by TFR and propose one policy lever each government has tried (or could try) to raise fertility.

Q14. Choose one of the three fertility-transition theories (Block 2) and explain, in 3-4 sentences, which one you find most convincing for explaining South Korea's current TFR, and why.

Q15. Design one additional survey question you would add to a Demographic and Health Survey to better distinguish whether low fertility in a given country is driven by Cm (marriage) or Cc (contraception) in the Bongaarts framework.

Q16. Drawing on Group Work 2 (Slides 51-53), explain why Hungary's and Singapore's pro-natalist spending has not restored replacement-level fertility, using at least one concept each from Becker and Easterlin.

Questions & Discussion

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